

Partical 5

varshithcomp1@VARSHITs-MacBook-Air ~ % nano \$Stask1.c
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LM #100 5.89 File: \$Stask1.c

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>
#include <string.h>

int main()
{
    int pipefd[2];
    pid_t pid;
    char buffer[100];

    int data[] = {10, 20, 30, 40, 50};
    int n = 5;

    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        exit(1);
    }

    pid = fork();

    if (pid < 0)
    {
        perror("fork");
        exit(1);
    }

    if (pid > 0)
    {
        close(pipefd[0]);

        printf("Producer (Parent):\n");
        for (int i = 0; i < n; i++)
        {
            sprintf(buffer, "%d", data[i]);
            printf("Producing: %d\n", data[i]);
            write(pipefd[1], buffer, strlen(buffer) + 1);
            usleep(100000);
        }

        close(pipefd[1]);
        wait(NULL);
    }
    else
    {
        close(pipefd[1]);

        printf("Consumer (Child):\n");
        while (read(pipefd[0], buffer, sizeof(buffer)) > 0)
        {
            printf("Consumed: %s\n", buffer);
        }

        close(pipefd[0]);
    }

    return 0;
}
```

Get Help
Exit

WriteOut
Justify

Read File
Where is

Prev Pg
Next Pg

Cut Text
UnCut Text

Cut Pos
To Spell

```
varshithcomple@VARSHITs-MacBook-Air ~ % nano S5task1.c
varshithcomple@VARSHITs-MacBook-Air ~ % nano S5task1.c
varshithcomple@VARSHITs-MacBook-Air ~ % gcc S5task1.c -o S5task1
clang: warning: overriding deployment version from '10.0' to '20.0' [-Woverriding-deployment-version]
varshithcomple@VARSHITs-MacBook-Air ~ % ./S5task1
Producer (Parent):
Producing: 10
Consumer (Child):
Consumed: 10
Producing: 20
Consumed: 20
Producing: 30
Consumed: 30
Producing: 40
Consumed: 40
Producing: 50
Consumed: 50
varshithcomple@VARSHITs-MacBook-Air ~ % █
```

TASK 2

UN PICO 8.09File: SStask2.c

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int pipefd[2];
    pid_t pid1, pid2;

    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        exit(1);
    }

    // First Child
    pid1 = fork();

    if (pid1 == -1)
    {
        perror("fork");
        exit(1);
    }

    if (pid1 == 0)
    {
        close(pipefd[0]);
        dup2(pipefd[1], STDOUT_FILENO);
        close(pipefd[1]);

        execlp("ls", "ls", "-l", (char *)NULL);

        perror("execlp ls");
        exit(1);
    }

    // Second Child
    pid2 = fork();

    if (pid2 == -1)
    {
        perror("fork");
        exit(1);
    }

    if (pid2 == 0)
    {
        close(pipefd[1]);
        dup2(pipefd[0], STDIN_FILENO);
        close(pipefd[0]);

        execlp("grep", "grep", ".c", (char *)NULL);

        perror("execlp grep");
        exit(1);
    }

    // Parent
    close(pipefd[0]);
    close(pipefd[1]);

    waitpid(pid1, NULL, 0);
    waitpid(pid2, NULL, 0);

    printf("Command executed successfully.\n");
}
```

Get Help

WriteOut

Read File

Prev Pg

Out Text

Our Pos

Exit

Justify

Where is

Next Pg

UnOut Text

To Spell

varshithcomple@VARSHITHs-MacBook-Air ~ % nano SStask2.c

varshithcomple@VARSHITHs-MacBook-Air ~ % gcc SStask2.c -o SStask2

clang: warning: overriding deployment version from '16.0' to '26.0' [-Woverriding-deployment-version]

varshithcomple@VARSHITHs-MacBook-Air ~ % ./SStask2

drwx----- 7 varshithcomple staff 224 1 Jun 14:38 Applications

drwx----- 17 varshithcomple staff 544 12 Sep 07:04 Desktop

drwx----- 8 varshithcomple staff 256 11 Sep 12:31 Documents

drwx----- 17 varshithcomple staff 544 12 Sep 06:48 Downloads

-rw-r--r-- 1 varshithcomple staff 1090 25 Jul 09:11 first.c

-rw-r--r-- 1 varshithcomple staff 934 28 Aug 13:48 fork.c

-rw-r--r-- 1 varshithcomple staff 992 28 Aug 13:54 fix.c

drwx----- 99 varshithcomple staff 3168 23 Jul 12:13 Library

drwx----- 5 varshithcomple staff 168 5 Jun 08:55 Movies

-rwxr-xr-x 1 varshithcomple staff 33448 19 Aug 07:47 ms

-rw-r--r-- 1 varshithcomple staff 1044 19 Aug 07:47 ms.c

drwx----- 5 varshithcomple staff 168 18 Feb 2026 Music

-rwxr-xr-x 1 varshithcomple staff 33544 19 Aug 07:27 notes

drwxr-xr-x 15 varshithcomple staff 488 28 Aug 14:05 OSSP

-rwxr-xr-x 1 varshithcomple staff 34088 28 Aug 15:11 parser

-rw-r--r-- 1 varshithcomple staff 2679 28 Aug 15:14 parser.c

drwx----- 6 varshithcomple staff 192 1 Jun 14:32 Pictures

-rwxr-xr-x 1 varshithcomple staff 33768 12 Sep 06:55 ptask

-rw-r--r-- 1 varshithcomple staff 964 12 Sep 06:53 ptask.c

-rwxr-xr-x 1 varshithcomple staff 33768 12 Sep 06:08 ptask2

-rw-r--r-- 1 varshithcomple staff 1012 12 Sep 06:57 ptask2.c

drwxr-xr-x 5 varshithcomple staff 168 18 Feb 2026 Public

-rwxr-xr-x 1 varshithcomple staff 33944 12 Sep 07:03 SStask1

-rw-r--r-- 1 varshithcomple staff 1042 12 Sep 07:00 SStask1.c

-rwxr-xr-x 1 varshithcomple staff 33768 12 Sep 07:00 SStask2

-rw-r--r-- 1 varshithcomple staff 1076 12 Sep 07:04 SStask2.c

-rwxr-xr-x 1 varshithcomple staff 33992 11 Aug 12:34 task1

-rw-r--r-- 1 varshithcomple staff 1468 11 Aug 12:37 task1.c

-rw-r--r-- 1 varshithcomple staff 1676 11 Aug 12:26 task1.c.save

-rwxr-xr-x 1 varshithcomple staff 34248 28 Aug 15:22 task2

-rw-r--r-- 1 varshithcomple staff 3403 28 Aug 15:22 task2.c

-rwxr-xr-x 1 varshithcomple staff 34136 28 Aug 15:27 task3

-rw-r--r-- 1 varshithcomple staff 2912 28 Aug 15:26 task3.c

Command executed successfully.

varshithcomple@VARSHITHs-MacBook-Air ~ %